

Meridian Fabrication Works -- Packaging & Ship Line

Visit date: 2026-09-10 Checklist: Theory of Constraints -- Find the Constraint

Demo data: this facility, its field notes, and its photos are entirely synthetic, generated by `scripts/generate_synthetic_data.py`. No real facility's data is used.

Executive Summary

2 critical/high-severity finding(s) from this theory of constraints -- find the constraint, led by 'Identify the Constraint' (6 occurrence(s)). 5 finding(s) total across Meridian Fabrication Works -- Packaging & Ship Line. Top priority: Stop-the-line priority -- address before the next shift starts. See the prioritized action list below for the full sequence.

(Synthesis method: template)

Findings

[CRITICAL] Identify the Constraint (x6)

6 instance(s) matched 'Identify the Constraint' (visible queue or wip buildup piling up in front of one station while other stations wait or run ahead). Field notes: There's a serious backlog in front of the shrink-wrap station, totes are piled up three deep waiting for it to catch up; Same thing an hour later, still a queue building up in front of shrink-wrap, hasn't cleared since this morning. Photo evidence: `b03_shrinkwrap_queue.png`, `b12_empty_conveyor.png`, `b16_endofshift_queue.png`.

Escalation: Escalated: this finding recurred across 6 separate notes (threshold: 3).

[HIGH] Exploit the Constraint (x2)

2 instance(s) matched 'Exploit the Constraint' (the busiest/slowest station sitting idle for a non-value-add reason -- changeover, break, waiting on a part -- while it is the pace-setter for the whole line). Field notes: Shrink-wrap machine went down for a changeover for almost twenty minutes right in the middle of the backlog, operators said they didn't have the next roll staged ahead of time; Another shrink-wrap stoppage, waiting on a part -- a replacement roll of film that wasn't staged nearby, probably ten minutes lost.

[MEDIUM] Subordinate Everything Else (x2)

2 instance(s) matched 'Subordinate Everything Else' (an upstream station producing faster than the constraint can consume, building excess wip that doesn't help overall output). Field notes: Meanwhile the labeling station upstream is running ahead, building a stack of labeled totes that

shrink-wrap can't keep up with, so they're just sitting there taking up floor space; Same thing at the case-pack step before that, running well ahead, building excess wip upstream that's just going to sit until shrink-wrap clears.

[MEDIUM] Elevate the Constraint (x2)

2 instance(s) matched 'Elevate the Constraint' (a visible capacity-adding opportunity at the constraint -- a second machine, an added shift, cross-training -- that hasn't been acted on). Field notes: Supervisor mentioned they've talked about adding a second shrink-wrap machine before the holiday peak but it hasn't been approved yet; One of the leads floated wanting to cross-train a picker to help run a second shrink-wrap head when volume spikes, but nobody's been trained yet.

[MEDIUM] Prevent Inertia / Constraint Has Moved (x1)

1 instance(s) matched 'Prevent Inertia / Constraint Has Moved' (signs the constraint has already shifted to a different station since the last review, but the team is still managing to the old one). Field notes: Talked to the shift lead, they said last month the real slow point was actually the case-erector -- used to be the bottleneck there -- but that's caught up now and shrink-wrap is the one falling behind.

Prioritized Action List

#	Finding	Severity	Recommended Action
1	Identify the Constraint	CRITICAL	Stop-the-line priority -- address before the next shift starts.
2	Exploit the Constraint	HIGH	Resolve within the week; assign an owner today.
3	Subordinate Everything Else	MEDIUM	Schedule a fix within the next 30 days.
4	Elevate the Constraint	MEDIUM	Schedule a fix within the next 30 days.
5	Prevent Inertia / Constraint Has Moved	MEDIUM	Schedule a fix within the next 30 days.

Methodology & Scope

Checklist: Theory of Constraints -- Find the Constraint (toc_constraint_walk). 13 voice note(s) and 3 photo(s) processed; 3 note(s)/photo(s) did not match any checklist item confidently and were excluded from findings rather than force-fit.

This checklist is swappable -- the same pipeline can run a Lean 5S audit, a Theory of Constraints constraint walk, or a safety & compliance walk from the same field-note and photo inputs, depending on which YAML config is selected (or recommended).

Appendix: Note-to-Finding Traceability

Note ID	Type	Matched Finding	Detail
B01	text	Identify the Constraint	There's a serious backlog in front of the shrink-wrap station, totes are piled up three deep waiting for it to catch up.
B02	text	Identify the Constraint	Same thing an hour later, still a queue building up in front of shrink-wrap, hasn't cleared since this morning.
B15	text	Identify the Constraint	Followed up at end of shift, the queue in front of shrink-wrap was still four totes deep, hadn't cleared by end of day.
B03	photo	Identify the Constraint	Pallets of totes queued up directly in front of the shrink-wrap machine, stacked higher than the machine itself.
B12	photo	Identify the Constraint	Empty conveyor section between the label station and shrink-wrap, no product moving through, consistent with operators' reports of waiting o
B16	photo	Identify the Constraint	End-of-shift photo showing totes still queued at the shrink-wrap infeed.
B04	text	Exploit the Constraint	Shrink-wrap machine went down for a changeover for almost twenty minutes right in the

			middle of the backlog, operators said they didn't have
B05	text	Exploit the Constraint	Another shrink-wrap stoppage, waiting on a part -- a replacement roll of film that wasn't staged nearby, probably ten minutes lost.
B06	text	Subordinate Everything Else	Meanwhile the labeling station upstream is running ahead, building a stack of labeled totes that shrink-wrap can't keep up with, so they're
B07	text	Subordinate Everything Else	Same thing at the case-pack step before that, running well ahead, building excess wip upstream that's just going to sit until shrink-wrap cl
B08	text	Elevate the Constraint	Supervisor mentioned they've talked about adding a second shrink-wrap machine before the holiday peak but it hasn't been approved yet.
B09	text	Elevate the Constraint	One of the leads floated wanting to cross-train a picker to help run a second shrink-wrap head when volume spikes, but nobody's been trained
B10	text	Prevent Inertia / Constraint Has Moved	Talked to the shift lead, they said last month the real slow point was actually the case-erector -- used to be the bottleneck there -- but t